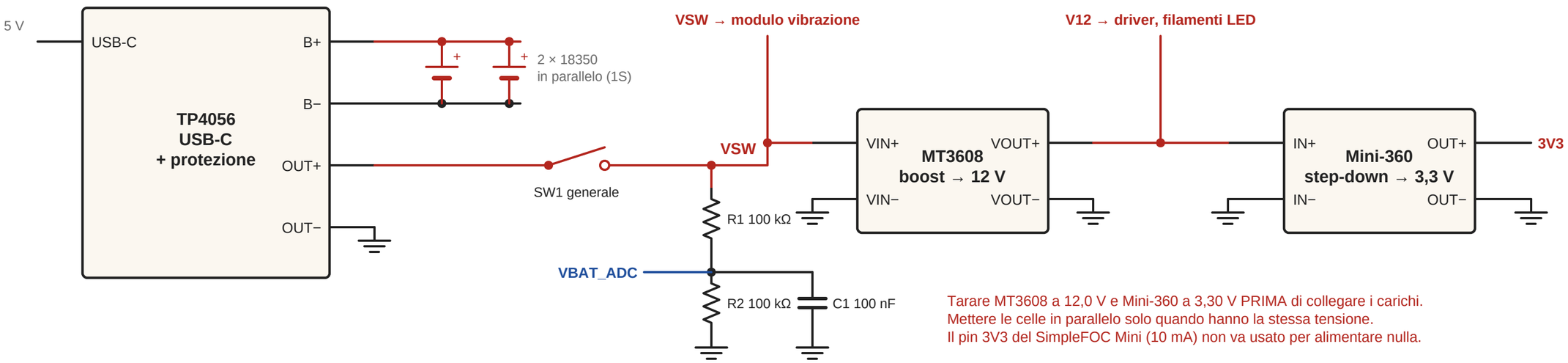
METRONOMUS · schema elettrico

rev. 0.1 · 17/10/2026 · bozza da verificare su breadboard · le etichette uguali indicano piedini collegati tra loro

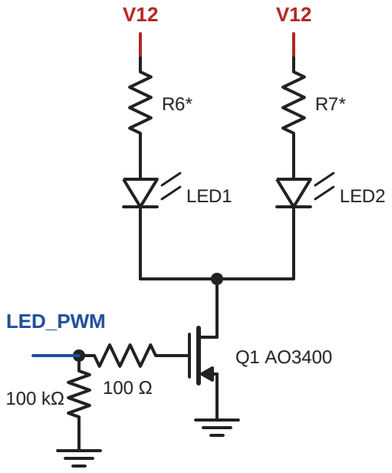
ALIMENTAZIONE



Tarare MT3608 a 12,0 V e Mini-360 a 3,30 V PRIMA di collegare i carichi.
Mettere le celle in parallelo solo quando hanno la stessa tensione.
Il pin 3V3 del SimpleFOC Mini (10 mA) non va usato per alimentare nulla.

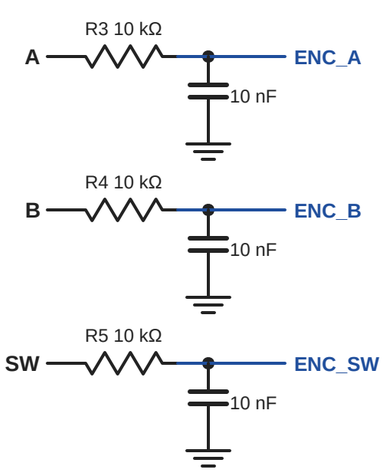
CONTROLLO E ATTUATORI

Filamenti LED (×2) sul pendolo

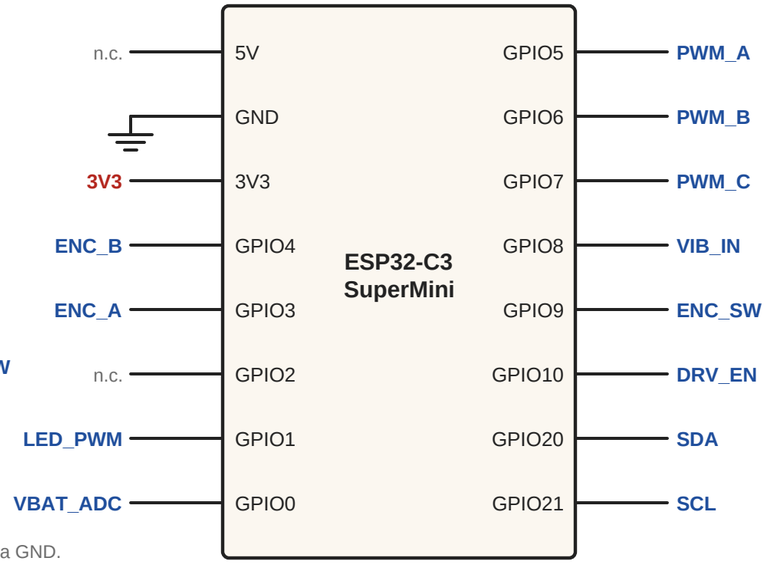


* R6, R7 = (12 V − Vf) / I, da calcolare dopo aver misurato la Vf del filamento (≈ 300 Ω, 0,5 W). Il filo di ritorno corre lungo il filamento.

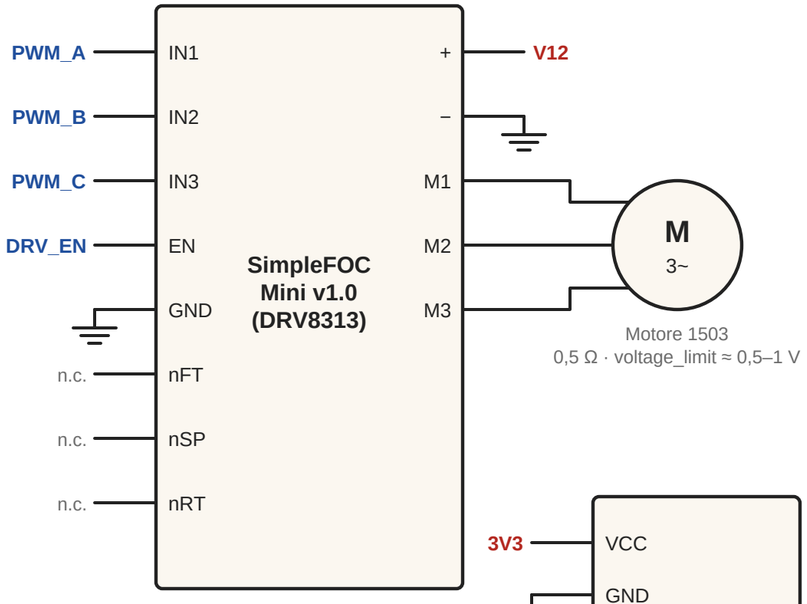
Encoder EC11 + antirimbalzo



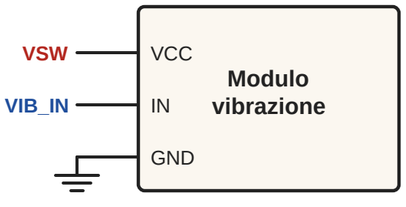
C (comune) e secondo piedino del pulsante a GND. Pull-up interne dell'ESP32 attive.

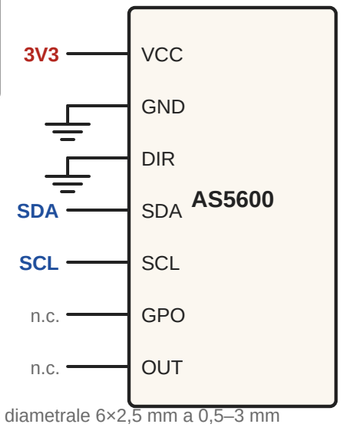


GPIO9 = tasto BOOT · GPIO8 = LED di bordo
GPIO2 libero (riserva: nFT del driver)



Motore 1503
0,5 Ω · voltage_limit ≈ 0,5–1 V





47–100 µF + 100 nF su VCC · impulsi ≥ 60 ms · magnete AS5600 diametrale 6×2,5 mm a 0,5–3 mm